

RESCUE JUNIOR - Bill of Materials (BOM)

Team name: TOKAI Ghost

Every Line/Maze team has to submit a bill of materials for their robot

Instructions:

*Enter Team name above
*All costs need to be in local currency and their approximate conversion into dollars.
*All components worth less can be summarized in one line quantity 1 and their total cost (e.g. Screws)
*In the Local Currency Name column, the team must enter the name of the local currency, if it is different from US dollars.
*In the software sheet, in the Author column, the name of the designer, writer, or company that owns the software or library should be entered. If the team created their own algorithm, dataset, or AI model, they can include their names as authors.
*On the hardware sheet, in the "Author" column, enter the name of the company, engineer, or part manufacturer. If the team made a custom part, such as the robot housing, the team should include their names as the authors.

IMPORTANT The "Hardware" sheet is not required for simulation

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Name of the Local Currency	Total Cost Local Currency	Total Cost (\$) U.S.A. Dollars
Japanese Yen JPY	¥69,702	\$434.48

#	Component	Part name	Autor	Source	Quantity	Unit cost (Local Currency)	Unit cost (US Dollars)	Total Cost (Local Currency)	Total Cost (USA Dollars)
1	Main Microcontroller	Teensy 4.0	PJRC	digiquey.jp	1	¥4,285	\$26.71	¥4,285	\$26.71
2	Image Processing SBC	Raspberry Pi 5 (8GB)	Foundation	akizukidenshi.com	1	¥17,600	\$109.70	¥17,600	\$109.70
3	Sub Microcontroller	Seeed XIAO RP2040	Seeed Studio	akizukidenshi.com	2	¥980	\$6.11	¥1,960	\$12.22
4	Camera Module	IMX219-120	Waveshare	waveshare.com	1	¥2,700	\$16.83	¥2,700	\$16.83
5	Line Sensor	LBR-127HLD	Letex Technology	akizukidenshi.com	16	¥75	\$0.47	¥1,200	\$7.52
6	Color Sensor	S9706	Hamamatsu Photonics	akizukidenshi.com	2	¥600	\$3.74	¥1,200	\$7.48
7	ToF Sensor Module	Custom VL53L0X Module	Takumi / JLCPCB	jlcpcb.com	6	¥788	\$4.91	¥4,728	\$29.46
8	Micro Switch	SS-10GL13	OMRON	akizukidenshi.com	2	¥160	\$1.00	¥320	\$2.00
9	Tactile Switch	DTS-63-N-V	Cosland	akizukidenshi.com	2	¥15	\$0.09	¥30	\$0.18
10	RGB LED	WS2812B	Worldsemi	akizukidenshi.com	6	¥50	\$0.31	¥300	\$1.86
11	IMU	BNO055	Bosch Sensortec	akizukidenshi.com	1	¥2,450	\$15.27	¥2,450	\$15.27
12	Display	SSD1306 128x64	Electronics	akizukidenshi.com	1	¥580	\$3.61	¥580	\$3.61
13	Drive Servo Motor	STS3032	Feetech Robotics	akizukidenshi.com	4	¥4,800	\$29.92	¥19,200	\$119.68
14	Rescue Servo Motor	SG90	TowerPro	akizukidenshi.com	2	¥650	\$4.05	¥1,300	\$8.10
15	Rescue Servo Motor	SG92R	TowerPro	akizukidenshi.com	3	¥780	\$4.86	¥2,340	\$14.58
16	Camera Servo Motor	GH-S37D	Generic	GH-S37D	1	¥580	\$3.62	¥580	\$3.62
17	Main PCB	Custom made	Takumi / JLCPCB	jlcpcb.com	1	¥5	\$0.03	¥5	\$0.03
18	Line Sensor PCB	Custom made	Takumi / JLCPCB	jlcpcb.com	1	¥5	\$0.03	¥5	\$0.03
19	Power PCB	Custom made	Takumi / JLCPCB	jlcpcb.com	1	¥20	\$0.12	¥20	\$0.12
20	Rescue PCB	Custom made	Takumi / JLCPCB	jlcpcb.com	1	¥1	\$0.01	¥1	\$0.01
21	Front PCB	Custom made	Takumi / JLCPCB	jlcpcb.com	1	¥2	\$0.01	¥2	\$0.01
22	Side PCB	Custom made	Takumi / JLCPCB	jlcpcb.com	2	¥1	\$0.01	¥2	\$0.02
23	Frame Metal Parts	Custom made	Takumi / JLCPCB	jlcpcb.com	1	¥30	\$0.19	¥30	\$0.19
24	Metal Gear (Motor Side)	Custom made	Takumi / JLCPCB	jlcpcb.com	4	¥19	\$0.12	¥76	\$0.48
25	Metal Gear (Wheel Side)	Custom made	Takumi / JLCPCB	jlcpcb.com	4	¥32	\$0.20	¥128	\$0.80
26	Main Chassis Components	Custom made	Takumi	3D Printer	1	¥2,400	\$14.96	¥2,400	\$14.96
27	Wheels	Custom made	Takumi	Silicone Rubber Molding	4	¥90	\$0.56	¥360	\$2.24
28	Fasteners	Screws, nuts, spacers, etc.		wilco.jp	1	¥2,200	\$13.71	¥2,200	\$13.71
29	Wiring	Cables, connectors, etc.		akizukidenshi.com	1	¥1,200	\$7.48	¥1,200	\$7.48
30	Li-Po Battery	11.1V 2200mAh 30C	Fullymax Battery	goohobby.com	1	¥2,500	\$15.58	¥2,500	\$15.58

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Name of the	Total Cost	Total Cost (\$)
Local Currency	Local Currency	U.S.A. Dollars
Japanese Yen JPY	¥0.00	\$0.00

#	Name	Software's Tool/Library	Description	Source	Autor	Currency)	Dollars)
1	Open CV	Library	Image processing library	OpenCV - Biblioteca abierta de C	Opencv.org	FREE	FREE
2	Numpy	Library	Numerical computing library	NumPy	Numpy developers	FREE	FREE
3	Pyserial	Library	Serial communication library	https://pythonhosted.org/pyserial/	Pyserial developers	FREE	FREE
4	PyTorch	Library	Deep learning framework	https://pytorch.org/	Meta Platforms	FREE	FREE
5	Picamera2	Library	Camera control library for Raspberry Pi	Picamera2 Docs	Raspberry Pi Foundation	FREE	FREE
6	ultralytics	Library	YOLO-based object detection library	https://docs.ultralytics.com/	Ultralytics	FREE	FREE
7	Albumentations	Library	Image augmentation library	https://albumentations.ai/	Albumentations Team	FREE	FREE
8	Adafruit SSD1306	Library	OLED display driver library	https://docs.arduino.cc/libraries/ad	Adafruit Industries	FREE	FREE
9	Adafruit GFX Library	Library	Graphics rendering library	https://docs.arduino.cc/libraries/ad	Adafruit Industries	FREE	FREE
10	Adafruit BNO055	Library	IMU sensor interface library	https://docs.arduino.cc/libraries/ad	Adafruit Industries	FREE	FREE
11	Adafruit NeoPixel	Library	RGB LED control library	https://docs.arduino.cc/libraries/ad	Adafruit Industries	FREE	FREE
12	SCServo	Library	Serial servo motor control library	https://docs.arduino.cc/libraries/sc	Feetech Robotics	FREE	FREE
13	pololu VL53L0X	Library	Time-of-flight distance sensor library	https://github.com/pololu/vl53l0x-z	Pololu Corporation	FREE	FREE
14	Visual Studio Code	Software's Tool	Software	Visual Studio Code: edición de cód	Microsoft	FREE	FREE
15	Autodesk Fusion	Software's Tool	3D CAD software	Fusión de Autodesk Software ba	Autodesk	FREE/EDUCATION	FREE/EDUCATION
16	Git	Software's Tool	Version control system	Git - Descargas	Conservancy	FREE	FREE
17	PlatformIO IDE	Software's Tool	Data annotation tool	PlatformIO	PlatformIO Labs	FREE	FREE
18	Label Studio	Software's Tool	Image annotation tool	https://labelstud.io	HumanSignal	FREE	FREE
19	KiCad	Software's Tool	PCB design software	Kicad	KiCad Developers Team	FREE	FREE